Shova Rani

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Education

Bachelor of Science in Civil Engineering — Jan. 2022–Present
Bangladesh University of Engineering and Technology (BUET) — Dhaka, Bangladesh
Student ID-2004117, *Level*-4, *Term*-1, *CGPA*-3.26/4.00
Major-Geotechnical Engineering, Minor-Environmental Engineering

Tentative Time of Graduation- November, 2026

Higher Secondary School Certificate — July 2018–July 2020
Brahmanbaria Govt. College (BGC) — Brahmanbaria, Bangladesh
Group-Science, GPA-5.00/5.00

Secondary School Certificate — 2018–2020
Govt. Model Girls' High School (GMGHS) — Brahmanbaria, Bangladesh
Group-Science, GPA-5.00/5.00

Skills

Software & Programming Language: Arc-GIS, Python, SAP, ETABS, Autocad, Microsoft Office, Matlab, html, C, C++, LaTeX
Languages: Bangla and English
Others: Anchoring, Extempore Speech, Debating, Public Speaking, Writing, Organizing, Management

Research Experiences

Research Assistant — *02/08/2026-Present*

- Under Dr. Md. Ferdous Alam, Associate Professor, Dr. Md. Azijul Islam, Assistant Professor, Mirza Md. Nazmus Sakib, Assistant Professor, Geotechnical Engineering Laboratory, Bangladesh University of Engineering and Technology (BUET)

Reviewer — *February 2026*

- NSF NHERI GSC 4th International Mini-Conference | Remote

Undergraduate Research Assistant — *14/07/2024- 15/06/2026*

- Under Dr. Mohammad Shariful Islam, Professor, Department of Civil Engineering, Bangladesh University of Engineering and Technology (BUET)

Undergraduate Research Assistant — *14/07/2024- 15/06/2026*

- Under Shantanu Paul, Assistant Professor, Department of Civil Engineering, Bangladesh University of Engineering and Technology (BUET)

Undergraduate Research Assistant — *14/07/2024- 15/06/2026*

- Under Mohammad Ziaul Hoque, Graduate Student, Department of Civil Engineering, Bangladesh University of Engineering and Technology (BUET)

Engineer Intern — *November 2025*

- Housing and Building Research Institute (HBRI)| Full time.

Research Paper Publications

Under Review | 8th International Conference on Advances in Civil Engineering (ICACE-2026) — *December 2026*

- Spatiotemporal Variation of Water Quality Parameters in Hatirjheel Lake, Dhaka, Bangladesh: A Data Synthesis Approach

Under Review | Geosynthetics Conference 2027 — *February 2027*

- Shear Strength Properties of MICP-Treated Sand Reinforced with Discrete Nylon Fibers

Conference Paper | International Conference on Civil Engineering Research & Innovations (ICCEI 2025) — *December 2025*

- Evaluating Waste-Based Stabilizers: Review on Mechanical and Durability Performance for CSEB Production

Magazine Article | *Yearly Magazine of ASCE Student Chapter, BUET* *June 2024*

- On Stability Analysis and Erosion Control Using Vetiver Grass: A Green Approach

Research Poster Publications

NSF NHERI GSC 4TH INTERNATIONAL MINI-CONFERENCE 2026 *May 2026*

- Impact of Urbanization on Local Hydrology: A Computational Approach for Stormwater Risk Assessment in Dhanmondi, Dhaka

Professional Practices and Communication Sessional *February 2027*

- Solar Roadways: Paving the Way to Sustainable Energy

Architectural, Engineering and Planning Appreciation Sessional *February 2023*

- Modern Architecture from Civil Engineering Perspective

Canvas of Ideas| ACI BUET Student Chapter

- Evaluation of Compressive Strength of Concrete using Machine Learning

Workshop

BoQ Preparation *February 2024*

- Member of BOQ preparing team as Procuring Entity of a 3-storied mosque at Professional Practice and Communication Sessional.

Seminar Participant *February 2024*

- Seminar on High-Performance Concrete Using Microsilica by Mr. Surendra Rampyar Sharma organized by ACI BUET Student Chapter at BUET.

Co-Curricular Activities

Deputy Secretary (Content Development) June 7, 2024–May, 2025
ASCE BUET Student Chapter *Dhaka, Bangladesh*

Professional Experience

Engineer Intern November 2025
Housing and Building Research Institute (HBRI) · Full-time *Mirpur, Dhaka, Bangladesh (Onsite)*

- Mathematics Instructor of Class 9-10.
- Science Instructor of Class 8.

Academic Instructor March 2018–June 2018
The New Oxford Coaching Centre *Brahmanbaria, Bangladesh*

- Mathematics Instructor of Class 9-10.
- Science Instructor of Class 8.

Awards and Honors

Champion | Long Jump | BUET Annual Sports January 2026
Winner | Professor Fahima Khatun *Scholarship-2024* | By Vasha O Shahitto Anushilon Kendro | Brahmanbaria 27 April 2024
2nd Runner Up | High Jump | BUET Annual Sports January 2024
Honor | Onkur Begum Rokeya Recognition Award 2023 | By Ankur International 16 December 2023
Winner | R. A. M. Obaidul Muktadir Chowdhury *Scholarship-2023* | By Vasha O Shahitto Anushilon Kendro | Brahmanbaria 14 January 2023
Yellow Belt Certification | Taekwondo | Organized by BUET Self Defense Club September 2023
Winner August 2023

- Roadshow of Over The Wall: Season 2 by Marico Bangladesh Limited in collaboration with BUET Career Club.

Winner | R. A. M. Obaidul Muktadir Chowdhury Scholarship-2023 | By Vasha O Shahitto Anushilon Kendro — 14 January 2023

Gorobini Maa Award-2022 | By Universal Medical College and Hospital — 8 May 2022

- My mother received this Award for my academic excellence despite having socio-economic hardship.

Winner | Ghuddi Foundation's Education For All Project's Admission Coaching Scholarship Category 'A' — Jan. 2020

- Covered admission coaching fees and provided 4-month accommodation support for educational pursuits.

Winner & Best Speaker | Kendrobindu Intra-Batch & Inter-Batch Debate 2018 | Organized by Kendrobindu Academic Care — 2018

Adamya Medhabi Award | By Brac Bank-Prothom Alo Trust — September 2018

- This award, along with a monthly scholarship of 2 years, is given to help the meritorious students from financially challenged families.

Consistent Attendance Recognition Award | By GMGHS. — 2017

Winner | Hosne Ara Scholarship 2018, Brahmanbaria, Bangladesh.

Subdistrict Level Champion | Creative Talent Hunt by Government of Bangladesh — 2017

At Bangladesh Studies and Liberation War Segment — *Brahmanbaria, Bangladesh*

- Bangladesh Studies and Liberation War Segment consisted of 3 segments: Creative Question Answering, MCQ, Viva.

Best Reader Award | Bishwo Shahitto Kendro — 2017

Best Speaker | Intra-Class Debate | At GMGHS. — 2017

Champion | Extemptore Essay Writing | Organized by Public Library — 2017

1st Runner Up | Book Reading Assessment | Organized by Public Library — 2017

- On Hajar Bachhor Dhore Bangladeshi Bengali-Language Film.

1st Runner Up | Balance Race | Annual Sports Competition | At GMGHS — Jan. 2017

1st Runner Up | Quiz Contest | Organized by Vasha O Shahitto Anushilon Kendro, Brahmanbaria, Bangladesh — 2017

2nd Runner Up | Quiz Contest | Organized by Vorer Pakhi, Brahmanbaria, Bangladesh. — 2014

1st Runner Up | Art Competition | Organized by New Oxford Coaching Centre, Brahmanbaria, Bangladesh. — 2013

Projects

Capstone Project | Team Leader — *April 2025 - June 2026*

- Redevelopment of Dhaka's Nilkhet Bookshop

Professional Practice and Communication Sessional Project Proposal — *February 2024*

- Project Proposal on ECOSCAPE, Transforming Urban Living With Green Roof.

Computation Sessional Project | *Using MATLAB* — *September 2023*

- Prediction of Impact of Urbanization on Local Hydrology (Flood Prediction Tool).

Quantity Surveying Project — *September 2023*

- Materials And Cost Estimation of 7 Storied Residential Building.

Leadership and Volunteering

Graduated Alumni | Aspire Leaders program' 24 | Aspire Institute — August 7, 2024–November, 2024

- A fully funded leadership develpoment program founded by Harvard Business School Professors.

Volunteer — March 2023–Present

- Aunkur International

Volunteer | Ghuddi Foundation — Jan. 2020–Present

- Head Examiner of Scholarship Eligibility Test Exam Of 10,000 Students of Dhaka Centre-2021.
- Member of Viva Panel Committee-2021.
- Head Examiner of Scholarship Eligibility Test Exam of 10,000 Students Of Brahmanbaria Centre-2022.
- Organizer of a number of Counseling Programs for the Selected Scholarship Recipient.

Assistant Election Commissioner 2017

- Student Cabinet Elections at GMGHS

Representative July 2018–March 2020

- Kendrobindu Academic Care, Brahmanbaria, Bangladesh

Captain Jan. 2017–Jan. 2018

- At GMGHS

Leader | Developing the Reading Habit Programme

- Conducted by the SEQAEP Project under the Ministry of Education, Government of the People's Republic of Bangladesh

Leader | Red Crescent Parade Team Jan. 2012–March 2015
At GMGHS *Brahmanbaria, Bangladesh*

- Actively engaged in parades during Victory Day, Independence Day, and Annual Prize Giving ceremonies.
- Participant in multiple seminars organized by Bangladesh Red Crescent Society.

Reference

Dr. Mohammad Shariful Islam
Professor
Department of Civil Engineering, BUET
msharifulislam@ce.buet.ac.bd

Dr. Md. Ferdous Alam
Associate Professor
Department of Civil Engineering, BUET
ferdous@ce.buet.ac.bd